Isosceles and equilateral triangles

Core Concepts: Isosceles and equilateral triangles

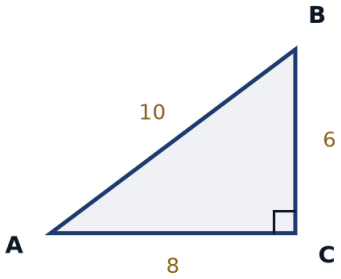
1 In $\triangle ABC$ studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 102°

b. 112°

c. 94°

d. 147°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

3 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 7

c. 5

d. 7

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95°

b. 105°

c. 87°

d. 147°

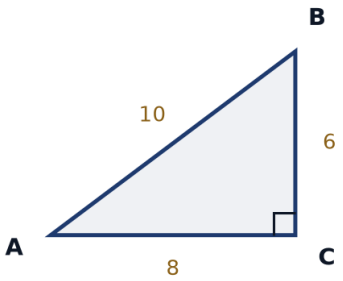
5 A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm

c. 28 cm

d. 63 cm



6 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 5

b. 9

c. 7

d. 9

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 88°

b. 98°

c. 80°

d. 147°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm

c. 35 cm

d. 112 cm

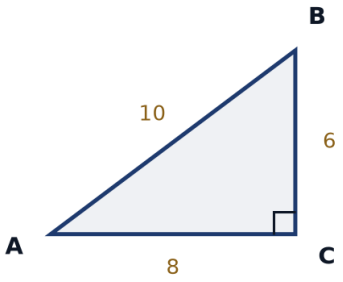
9 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 7

b. 11

c. 9

d. 11



Exam-Style Problem Solving & Modeling

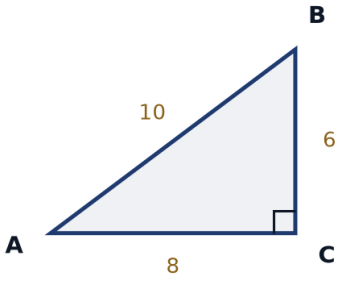
10 In $\triangle ABC$ studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 81°

b. 91°

c. 73°

d. 147°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm

c. 42 cm

d. 175 cm

12 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 9

b. 13

c. 11

d. 13

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